

DAVIDE ITALO SERRAMAZZA

1 ROEBUCK HOUSE, Dublin 14, Dublin, Ireland, D14PH27

+39 331 569 7012 / +353 892 473 441

davide.serramazza@gmail.com — <https://davide-serramazza.github.io/>

github.com/davide-serramazza — [linkedin.com/in/davide-italo-serramazza-88b9501a2/](https://www.linkedin.com/in/davide-italo-serramazza-88b9501a2/)

Professional Summary

PhD in Explainable AI for Time Series (thesis submitted; viva pending), with 6+ years of experience building and shipping ML systems. Created tsCaptum (PyPI) and drXAI (GitHub), reducing computational time by 1-2 orders of magnitude. Proven ability to bridge academic research and production-ready code. Seeking Applied Scientist or ML Engineer roles.

Technical Skills

- **Programming Languages:** Python, Nodejs, MATLAB, Java, JavaScript, C++, C, Bash script
- **ML/DL frameworks:** PyTorch, TensorFlow, Sktime, Aeon, Scikit-learn, Captum
- **XAI:** SHAP, KernelSHAP, LIME, Feature Ablation, dCAM
- **MLOps:** Docker, Git, CI/CD (familiar), MLflow (learning), Airflow (learning)
- **GenAI & Vector DB:** Actively exploring LLMs, RAG, vector databases (Weaviate)
- **Other:** Linux, MongoDB, FFmpeg, REST APIs, CUDA

Experience/Projects

University College Dublin

September 2022 – September 2026 (thesis submission)

PhD in AI for Time Series

Dublin, Ireland

- Designed and implemented interpretable ML systems for complex, sequential data, with a focus on feature analysis, model validation, and performance optimization.
- Built **tsCaptum** (PyPI), a Python library bridging XAI with time series frameworks, reducing computational time by 1-2 orders of magnitude without accuracy degradation
- Developed **drXAI**, an XAI-driven framework for feature selection and data reduction (ECML-PKDD 2026)
- Published **5 first-author papers** on XAI for time series classification <https://scholar.google.com/citations?user=DFHxPqMAAAAJ>

Università di Pisa / Mediateca Toscana

May 2020 – May 2022

Assegno di Ricerca (Research job)

Tuscany, Italy

- Built end-to-end ML pipeline: data ingestion → model training → deployment (API + database)
- Processed 100+ hours of documentary clips using deep learning algorithms
- *Technical stack:* Python, PyTorch, TensorFlow, GitHub, Node.js, MongoDB, FFmpeg, REST APIs

Education

PhD in XAI for Time Series

September 2022 – September 2026

University College Dublin (UCD)

Dublin, Ireland

Master in AI (School of Computer Science), Grade 110/110

September 2017 – March 2020

Università di Pisa

Pisa, Italy

Bachelor in Computer Science

September 2013 – September 2017

Università di Pisa

Pisa, Italy

Languages

- **Italian** (native)
- **English** (Professional working proficiency)